

1. Three small contracts

Mosaic gives authors three knobs for line breaks: a non-breaking space, a forced line break, and a soft hyphen. They are small contracts. Each does one thing, and only one thing.

1.1. Non-breaking space (U+00A0)

A literal U+00A0 between two words tells the line-breaker that the pair must travel together. The greedy breaker honours this because it splits only at ASCII whitespace, and U+00A0 is not ASCII whitespace.

Authors who refuse to break "Mr. Smith" across a line write the literal source `Mr. <NBSP>Smith` (where `<NBSP>` is one U+00A0 code point, two bytes in UTF-8). Rendered, the pair joins as Mr. Smith, visually indistinguishable from a regular space, but treated by the breaker as an indivisible unit. Likewise `page<NBSP>4` keeps the label and number together, rendered as page 4.

A single regular space, like the one between this word and the next, is, of course, breakable like any other.

1.2. Hard line break (\\)

The escape `\\` forces a line break inside a paragraph without the extra leading that a blank-line paragraph break would emit. Use it when the *line* is the unit of meaning: addresses, verse, captions, the heroic-but-doomed code path where you really do want this on one line

and this on the next without spawning a new paragraph.

Stack two of them

and you get a blank line between two otherwise-adjacent runs, still inside the same paragraph. The paragraph's overall spacing-after is paid once, at the end.

A lone trailing `\` (no second backslash) is not an escape; it emits the diagnostic W025 and falls through as a literal backslash character.

1.3. Soft hyphen (\- or U+00AD)

The soft hyphen marks a permitted-but-not-required break inside a word. It is the author saying "if you ever need to break this word, here is where to do it; otherwise pretend I was never here". The shorthand `\-` expands to a literal U+00AD; the codepoint itself works too.

Soft hyphens are *invisible when they aren't needed*: the codepoint is stripped before shaping, so a word like *supercalifragilisticexpialidocious* that comfortably fits a line renders as *supercalifragilisticexpialidocious* the same as if none of the `\-` shorthands had been written.

When a word *would* overflow the line, the greedy breaker now picks the latest marked

position whose prefix-plus-visible-hyphen still fits the remaining space, emits `prefix-` at the end of the current line, and continues with the suffix as the next word. If even the shortest marked prefix can't fit an empty line, the existing oversize-cluster fallback chops the word at glyph boundaries instead.

Optimal (non-greedy) selection across a whole paragraph is left for the eventual Knuth-Plass line-breaker; the greedy slice here is the smallest change that turns the SHY markers into a visible feature.